# PRODUCT INFORMATION
# LEWATIT® S 1567

**Lewatit® S 1567** is a food grade, strongly acidic cation exchange resin with beads of uniform size (monodisperse) based an a styrene-divinylbenzene copolymer. **Lewatit® S 1567** is manufactured without the use of solvent.

The monodisperse beads are chemically and osmotically very stable, and they can effectively be disinfected for the drinking water processing. The optimized kinetics lead to an increased operating capacity compared to ion exchange resins with heterodisperse bead size distribution.

**Lewatit® S 1567** is especially applicable for:

» softening in special systems with regular disinfection
» softening of drinking water

**Lewatit® S 1567** is adding special features to the resin bed:

» high exchange flow rates during regeneration and loading
» a good utilization of the total capacity
» a low demand for rinse water
» homogeneous throughput of regenerants, water and solutions; therefore a homogeneous working zone
» nearly linear pressure drop gradient for the whole bed depth; therefore operation with higher bed depth is possible

The special properties of this product can only be fully utilized if the technology and process used correspond to the current state-of-the-art. Further advice in this matter can be obtained from Lanxess, Business Unit Liquid Purification Technologies.

## Common Description

| | |
|---|---|
| Delivery form | $Na^+$ |
| Functional group | Sulfonic acid |
| Matrix | Styrenic |
| Structure | Gel |
| Appearance | Dark brown |

## Specified Data

| | | | |
|---|---|---|---|
| Uniformity coefficient | | max. | 1.1 |
| Mean bead size | d50 | mm | 0.60 (+-0.05) |
| Total capacity (delivery form) | | min. eq/L | 1.8 |

## Typical Physical and Chemical Properties

| | | | |
|---|---|---|---|
| Bulk density for shipment | (+/- 5%) | g/L | 810 |
| Density | | approx. g/mL | 1.28 |
| Water retention (delivery form) | | approx. weight % | 44-50 |
| Stability pH range | | | 0-14 |
| Storage time (after delivery) | | max. years | 2 |
| Storage temperature range | | °C | -20 - +40 |

## Operation

| | | | |
|---|---|---|---|
| Operating temperature | | max. °C | 120 |
| Operating pH range | during exhaustion | | 5-8 |
| Bed depth for single column | | min. mm | 800 |
| Back wash bed expansion per m/h (20°C) | | % | 4 |
| Specific pressure loss kPa*h/m² (15°C) | | kPa*h/m² (15°C) | 1 |
| Max. pressure loss during operation | | kPa | 200 |
| Specific flow rate | | max. BV/h | 40 |

## Regeneration

| | | | |
|---|---|---|---|
| NaCl regeneration | concentration | approx. wt. % | 8-12 |
| NaCl regeneration | quantity co-current | min. g/L resin | 100 |
| NaCl regeneration | quantity counter-current | min. g/L resin | 70 |
| Regeneration contact time | | min. minutes | 20 |
| Slow rinse at regeneration flow rate | | min. BV | 2 |
| Fast rinse at service flow rate | | min. BV | 2 |

## Additional Information & Regulations

**Safety precautions**
Strong oxidants, e.g. nitric acid, can cause violent reactions if they come into contact with ion exchange resins.

**Toxicity**
The safety data sheet must be observed. It contains additional data on product description, transport, storage, handling, safety and ecology.

**Disposal**
In the European Community Ion exchange resins have to be disposed, according to the European waste nomenclature which can be accessed on the internet-site of the European Union.

**Storage**
It is recommended to store ion exchange resins at temperatures above the freezing point of water under roof in dry conditions without exposure to direct sunlight. If resin should become frozen, it should not be mechanically handled and left to thaw out gradually at ambient temperature. It must be completely thawed before handling or use. No attempt should be made to accelerate the thawing process.

**Packaging**
The experience has shown that the packaging stability for reliable resin containment is limited to 24 months under the storage conditions described above. It is therefore recommended to use the product within this time frame; otherwise the packaging condition should be checked regularly.

This information and our technical advice – whether verbal, in writing or by way of trials – are given in good faith but without warranty, and this also applies where proprietary rights of third parties are involved. Our advice does not release you from the obligation to check its validity and to test our products as to their suitability for the intended processes and uses. The application, use and processing of our products and the products manufactured by you on the basis of our technical advice are beyond our control and, therefore, entirely your own responsibility. Our products are sold in accordance with the current version of our General Conditions of Sale and Delivery.

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